

CS 378H Spring 2024 Linear Algebra Worksheet

Due Tuesday, January 23, 2024 at 1:59 PM

“There is hardly any theory which is more elementary [than linear algebra], in spite of the fact that generations of professors and textbook writers have obscured its simplicity by preposterous calculations with matrices.” –J. Dieudonne

Name: Arham Lodha

UT EID: arl3759

Questions begin on the next page. Please answer each question in the spaces provided; if you need extra room, you can attach extra blank pages at the end of the worksheet. The worksheet follows the standard class collaboration policy. You may ask other students, the instructor, or the TA for help, but **all work written on this worksheet must be entirely your own**. Do not simply copy somebody else’s answers. You may look at linear algebra textbooks, online tutorials, math.stackexchange, or any other resources to help you complete the worksheet. Do not worry if you cannot do the problems from memory alone! You are allowed, and expected, to look up any formulas or definitions as needed. The problems are not intended to be difficult or time-consuming to solve, but rather, to refresh your memory on basic linear algebra concepts, and to give you a self-check on the prerequisites you will need to succeed in this class. If you get stuck on any problem, the TA or instructor would be happy to help you during office hours.

Useful Formulas

- Dot product: $(u_x, u_y, u_z, u_w) \cdot (v_x, v_y, v_z, v_w) = u_x v_x + u_y v_y + u_z v_z + u_w v_w$.
- Cross product: $(u_x, u_y, u_z) \times (v_x, v_y, v_z) = (u_y v_z - u_z v_y, u_z v_x - u_x v_z, u_x v_y - u_y v_x)$.
- Norm: $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{\mathbf{u}^T \mathbf{u}}$.
- Unit vector: $\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$.
- Angles between vectors: $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$, $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$.

1. Recall that a function $f(\mathbf{v})$ of a vector $\mathbf{v}$ is *linear* if $f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$ and $f(\mathbf{v} + \mathbf{w}) = f(\mathbf{v}) + f(\mathbf{w})$. For each of the following functions, (a) state whether or not the function is linear, and (b) if the function is not linear, give a counterexample where the function violates one of the above properties. You do *not* need to provide a proof if the function is linear; simply stating it's linear is enough.
 - (a) (1 point) Adding a non-zero constant vector $\mathbf{c}$ to the input vector, $f(\mathbf{v}) = \mathbf{v} + \mathbf{c}$.
 - a) The function is not linear
 - b) $\forall \mathbf{v} \in V, f(0\mathbf{v}) = f(0) = 0 + \mathbf{c} = \mathbf{c} \neq 0f(\mathbf{v}) = 0(0\mathbf{v} + \mathbf{c}) = 0$
 - (b) (1 point) The constant function $f(\mathbf{v}) = 42$.
 - a) the function is not linear
 - b) $\forall \mathbf{v} \in V, f(0) = 42 \neq 0f(\mathbf{v}) = 0$
 - (c) (1 point) Swapping the coordinates of a 2D vector: $f(\mathbf{v}) = (\mathbf{v}_y, \mathbf{v}_x)$
 - a) The function is linear
 - (d) (1 point) Vector norm, $f(\mathbf{v}) = \|\mathbf{v}\|$.
 - a) The function is not linear.
 - b) Let $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and let $w = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. $f(v) = 1 = f(w)$. $f(v + w) = \sqrt{2} \neq f(v) + f(w) = 2$
 - (e) (2 points) The function returning the determinant of a 2×2 matrix: $f(M) = \det M$.
 - a) Not linear
 - b) $f(2I) = \det(2I) = 4 \neq 2f(I) = 2$
 - (f) (2 points) Cross product with a fixed vector $\mathbf{c}$: $f(\mathbf{v}) = \mathbf{v} \times \mathbf{c}$.
 - a) Function is linear
 - (g) (2 points) Angle $\theta(\mathbf{v}) = \angle(\mathbf{v}, \mathbf{c})$ between $\mathbf{v}$ and a fixed vector $\mathbf{c}$.
 - a) Function is not linear
 - b) $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, c = w = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \theta(v) = \frac{\pi}{2}, \theta(w) = 0. \theta(v + w) = \frac{\pi}{4} \neq \theta(v) + \theta(w) = \frac{\pi}{2}$

2. Although in practice you will not need to do a lot of linear algebra by hand (that's what computers are for!), you should be able to perform basic computations, and know how to use computational resources like Matlab, Mathematica, Wolfram Alpha to help you debug your code.

- (a) (3 points) A rotation matrix R is a square matrix satisfying the equations $R^T R = I$ and $\det R = 1$, where I is the identity matrix. Is the following matrix (approximately) a rotation matrix? How do you know? **You do not need to do the calculation by hand or show work.**

$$\begin{bmatrix} 0.743 & -0.669 \\ 0.669 & 0.743 \end{bmatrix}$$

Yes, it is approximately a rotation matrix because

$$\begin{bmatrix} 0.743 & -0.669 \\ 0.669 & 0.743 \end{bmatrix}^T \begin{bmatrix} 0.743 & -0.669 \\ 0.669 & 0.743 \end{bmatrix} = \begin{bmatrix} 99961/100000 & 0 \\ 0 & 99961/100000 \end{bmatrix} \approx I$$

- (b) (3 points) Does the matrix

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

have an inverse? How do you know for sure? If an inverse exists, compute it. **You do not need to do the calculation by hand or show work.**

No, M is not invertible because $\det(M) = 0$.

- (c) (4 points) Use your favorite computer linear algebra tool to solve

$$\begin{bmatrix} 2 & 1 & 7 & 8 \\ -1 & 1 & 2 & -3 \\ 1 & 3 & 5 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

for $\mathbf{v}$. Round each coordinate of your answer to the nearest 0.01. **You do not need to do the calculation by hand or show work.**

$$\mathbf{v} = \begin{bmatrix} -16.52 \\ -4.35 \\ 0.91 \\ 4 \end{bmatrix}$$

3. Let $\mathbf{c} \in \mathbb{R}^2$ be a fixed vector. Let

$$f(\mathbf{v}) = \left(I - 2 \frac{\mathbf{c}\mathbf{c}^T}{\mathbf{c}^T\mathbf{c}} \right) \mathbf{v},$$

for vectors $\mathbf{v} \in \mathbb{R}^2$ in the plane, where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity matrix.

- (a) (1 point) Compute $\mathbf{c}\mathbf{c}^T$ and $\mathbf{c}^T\mathbf{c}$ for $\mathbf{c} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Are these similar-looking expressions actually the same, or different?

$$\mathbf{c}\mathbf{c}^T = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$$

$$\mathbf{c}^T\mathbf{c} = \begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = 1 + 4 = 5$$

No, the similar looking expressions are not the same, for $c = \begin{bmatrix} 1 & -2 \end{bmatrix}$ the first expression is a 2×2 matrix while the second is a scalar.

- (b) (1 point) Is the function $f(\mathbf{v})$ linear? If not, provide a counterexample.

Yes it is linear. $\forall v, w \in \mathbb{R}^2$ and $\forall a, b \in \mathbb{R}$, $f(av + bw) = \left(I - 2 \frac{\mathbf{c}\mathbf{c}^T}{\mathbf{c}^T\mathbf{c}} \right) (av + bw) = a \left(I - 2 \frac{\mathbf{c}\mathbf{c}^T}{\mathbf{c}^T\mathbf{c}} \right) v + b \left(I - 2 \frac{\mathbf{c}\mathbf{c}^T}{\mathbf{c}^T\mathbf{c}} \right) w = af(v) + bf(w)$.

- (c) (2 points) Compute $f(\mathbf{c})$.

$$f(\mathbf{c}) = \left(I - 2 \frac{\mathbf{c}\mathbf{c}^T}{\mathbf{c}^T\mathbf{c}} \right) \mathbf{c} = \mathbf{c} - 2 \frac{\mathbf{c}\mathbf{c}^T\mathbf{c}}{\mathbf{c}^T\mathbf{c}} = -\mathbf{c}$$

- (d) (2 points) Compute $f(\mathbf{v})$, where $\mathbf{v}$ is any vector perpendicular to $\mathbf{c}$ (i.e. $\mathbf{v} \cdot \mathbf{c} = 0$).

$$f(v) = \left(I - 2 \frac{\mathbf{c}\mathbf{c}^T}{\mathbf{c}^T\mathbf{c}} \right) v = v - 2 \frac{\mathbf{c}\mathbf{c}^T v}{\mathbf{c}^T\mathbf{c}} = v - 2 \frac{\mathbf{c}(\mathbf{c} \cdot v)}{\mathbf{c}^T\mathbf{c}} = v - 2(0) = v$$

- (e) (4 points) Describe using words, and no equations, what the function f does to vectors. You may find it helpful to probe the behavior of $f(\mathbf{v})$ for concrete fixed vectors $\mathbf{c}$ and draw some pictures. Your answers to parts (c) and (d) should also prove useful.

From the input vector $v \in \mathbb{R}^2$, f subtracts twice the portion of v which is in the direction of c , in other words it inverts the portion of v which is in the direction of the fixed vector c . More simply, it reflects the input vector v with respect to the plane normal to c .

4. A key skill in linear algebra is turning systems of linear equations into matrix equations of the form $A\mathbf{x} = \mathbf{b}$, where A and $\mathbf{b}$ are known, and $\mathbf{x}$ is unknown.

- (a) (5 points) Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ and $\mathbf{w}$ be four different vectors in $\mathbb{R}^3$. You are told that

$$\mathbf{w} = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2 + \gamma\mathbf{v}_3$$

for some unknown real numbers α, β , and γ . Write a matrix equation (of the form $A\mathbf{x} = \mathbf{b}$) that would allow you to solve for these unknown scalars. (**You do not need to actually solve the equation.**)

$$A = \begin{bmatrix} | & | & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} v_{1x} & v_{2x} & v_{3x} \\ v_{1y} & v_{2y} & v_{3y} \\ v_{1z} & v_{2z} & v_{3z} \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$

$$A\mathbf{x} = \begin{bmatrix} | & | & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} v_{1x} & v_{2x} & v_{3x} \\ v_{1y} & v_{2y} & v_{3y} \\ v_{1z} & v_{2z} & v_{3z} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \mathbf{w}$$

- (b) (5 points) Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be three different vectors in $\mathbb{R}^3$, and k_1, k_2, k_3 three real numbers. You are told that

$$k_1 = \mathbf{w} \cdot \mathbf{v}_1$$

$$k_2 = \mathbf{w} \cdot \mathbf{v}_2$$

$$k_3 = \mathbf{w} \cdot \mathbf{v}_3$$

for some unknown vector $\mathbf{w} \in \mathbb{R}^3$. Write down a matrix equation that would allow you to solve for this vector $\mathbf{w}$.

$$A = \begin{bmatrix} v_{1_x} & v_{1_y} & v_{1_z} \\ v_{2_x} & v_{2_y} & v_{2_z} \\ v_{3_x} & v_{3_y} & v_{3_z} \end{bmatrix}$$

$$x = w$$

$$b = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}$$

$$Aw = \begin{bmatrix} v_{1_x} & v_{1_y} & v_{1_z} \\ v_{2_x} & v_{2_y} & v_{2_z} \\ v_{3_x} & v_{3_y} & v_{3_z} \end{bmatrix} w = b$$

5. Suppose you wrote a computer program that drew an animation by plotting the motion of a point with coordinates $\mathbf{p}(t)$ in the plane, as a function of time $t \in [0, \infty)$. Describe what you would see on the screen, as time evolves. (The intent is not for you to actually write such a program, but doing so is not forbidden if you're stuck.) If you sketch pictures, clearly indicate how the figure is changing over time.

(a) (5 points)

$$\mathbf{p}(t) = (1 - t) \begin{bmatrix} -3 \\ 4 \end{bmatrix} + t \begin{bmatrix} -3 \\ 5 \end{bmatrix}.$$

A point starting at $\begin{bmatrix} -3 & 4 \end{bmatrix}$ and going in the $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ direction in a straight line.

(b) (5 points)

$$\mathbf{p}(t) = \begin{bmatrix} \cos(t^2) & -\sin(t^2) \\ \sin(t^2) & \cos(t^2) \end{bmatrix} \begin{bmatrix} e^t \\ 0 \end{bmatrix}.$$

A point starting at $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and going around a spiral whose points are getting exponentially further from the origin with respect to time.